Variant 11 as a Minimal Nontrivial Instance of the Common Support Interval

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Abstract

This note places Variant 11,

$$f_{11}(p,q) = \neg q,$$

within the support structure subsequently established by the presentation-support framework and the Common Support Interval result. For the fixed frame $N = \{p, q\}$, the determinational support is

$$R_{f_{11}} = \{q\}.$$

The presentation-support result from III-A yields

$$\mathfrak{S}_N(f_{11}) = [\{q\}, \{p, q\}]_{\subseteq},$$

while the Common Support Interval result gives

$$\mathfrak{F}_N(f_{11}) = \mathfrak{S}_N(f_{11}).$$

Hence the factorization-support and presentation-support families of Variant 11 coincide and are given by

$$\mathfrak{F}_N(f_{11}) = \mathfrak{S}_N(f_{11}) = \{\{q\}, \{p, q\}\}.$$

The equality concerns support families and does not identify coordinate factorization with term presentation. Within the class of nonconstant finite typed Boolean functions with proper determinational support, Variant 11 attains the smallest possible ambient-frame size and is therefore a minimal nontrivial instance of the common support interval.

Keywords: finite typed Boolean functions, determinational support, syntactic support, coordinate factorization, term presentation, fictitious variables, common support interval.
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1 Purpose and Scope

The purpose of this note is to state precisely the structural position of Variant 11 [1] in light of the subsequent development of the presentation-support framework in III-A [2] and the Common Support Interval result [3].

The note does not revise the original Variant 11 paper, extend III-A, or extend the Common Support Interval theorem. It introduces no new theory of transformations and makes no claim of uniqueness or exceptional status for Variant 11.

The relevant genealogy is interpretive:

$$\text{Variant 11} \longrightarrow \text{III-A} \longrightarrow \text{Common Support Interval.}$$

This sequence records the order in which the structural position considered here becomes progressively explicit. It is not intended to suggest that the later general results are logical consequences of Variant 11.

Throughout the note, $\mathcal{L}$ denotes the fixed Boolean term language used in III-A and in the Common Support Interval result.

At the support level, the notation remains

$$R_f, \qquad S_\tau, \qquad \mathfrak{S}_N(f), \qquad \mathfrak{F}_N(f), \qquad [R_f, N]_\subseteq.$$

2 Variant 11 and Its Determinational Support

Let

$$N = \{p, q\},$$

and consider the finite typed Boolean function

$$f_{11} : \mathbf{2}^N \longrightarrow \mathbf{2}$$

defined by

$$f_{11}(p, q) = \neg q.$$

The two coordinates have different determinational roles. For every $q \in \mathbf{2}$,

$$f_{11}(0, q) = f_{11}(1, q) = \neg q,$$

so changing p while keeping q fixed never changes the value of the function. Thus p is fictitious relative to f_{11}.

By contrast, for every $p \in \mathbf{2}$,

$$f_{11}(p, 0) = 1, \qquad f_{11}(p, 1) = 0.$$

Hence changing q while keeping p fixed changes the value of the function, so q is real relative to f_{11}.

Therefore the determinational support of Variant 11 is

$$\boxed{R_{f_{11}} = \{q\}.}$$

Equivalently,

$$q \in R_{f_{11}}, \qquad p \in N \setminus R_{f_{11}},$$

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and hence

$$\boxed{R_{f_{11}} = \{q\} \subsetneq N = \{p, q\}.}$$

This proper inclusion concerns the relation between the typed function and its formal frame. In particular, the membership

$$p \in N$$

does not by itself imply

$$p \in S_\tau$$

for an arbitrary term presentation τ of f_{11}. Literal syntactic occurrence is presentation-dependent and is recorded separately by the syntactic support S_τ.

The possible syntactic supports of presentations of Variant 11 are determined in the next section.

3 Presentation Supports of Variant 11

For the fixed Boolean term language $\mathcal{L}$, III-A [2] establishes that the presentation-support family of a finite typed Boolean function satisfies

$$\mathfrak{S}_N(f) = [R_f, N]_\subseteq.$$

Applying this result to Variant 11, with

$$R_{f_{11}} = \{q\} \qquad \text{and} \qquad N = \{p, q\},$$

gives

$$\mathfrak{S}_N(f_{11}) = [\{q\}, \{p, q\}]_\subseteq.$$

Since there is no subset strictly between $\{q\}$ and $\{p, q\}$, it follows that

$$\boxed{\mathfrak{S}_N(f_{11}) = \Big\{\{q\}, \{p, q\}\Big\}.}$$

Thus both exact syntactic-support levels occur among term presentations of f_{11} in the fixed language $\mathcal{L}$. In particular, there exists a term presentation τ of f_{11} in $\mathcal{L}$ such that

$$S_\tau = \{p, q\} = N.$$

For such a presentation,

$$p \in S_\tau, \qquad p \notin R_{f_{11}}.$$

Hence the coordinate p may occur literally in a term presentation of f_{11} while remaining absent from its determinational support.

This gives the precise presentation-level form of the separation between literal syntactic occurrence and real determination:

$$\boxed{p \in S_\tau \setminus R_{f_{11}}}$$

for at least one term presentation τ of Variant 11.

The relation between the presentation-support and factorization-support families is considered in the next section.

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4 The Common Support Interval

The preceding section established, for Variant 11 in the fixed Boolean term language $\mathcal{L}$,

$$\mathfrak{S}_N(f_{11}) = [\{q\}, \{p,q\}]_{\subseteq} = \Big\{\{q\}, \{p,q\}\Big\}.$$

The Common Support Interval result [3] proves the equality of the presentation-support and factorization-support families of a finite typed Boolean function. Applied to the same function f_{11} on the same frame

$$N = \{p,q\},$$

it gives

$$\mathfrak{F}_N(f_{11}) = \mathfrak{S}_N(f_{11}).$$

Combining the two statements yields the common support interval of Variant 11.

Corollary 4.1 (Common support interval of Variant 11)**.** *For*

$$f_{11} : \mathbf{2}^{\{p,q\}} \longrightarrow \mathbf{2}, \qquad f_{11}(p,q) = \neg q,$$

one has

$$\boxed{\mathfrak{F}_N(f_{11}) = \mathfrak{S}_N(f_{11}) = [\{q\}, \{p,q\}]_{\subseteq} = \Big\{\{q\}, \{p,q\}\Big\}.}$$

Remark 4.2 (Structural distinction). The equality

$$\mathfrak{F}_N(f_{11}) = \mathfrak{S}_N(f_{11})$$

is an equality of support families. It does not identify coordinate factorization with term presentation. The two constructions remain structurally distinct, even though they admit exactly the same coordinate supports for the fixed typed function f_{11} on the fixed frame N.

Thus the two support levels

$$\{q\} \qquad \text{and} \qquad \{p,q\}$$

occur on both sides of the comparison: as factorization supports and as exact syntactic supports. Their coincidence is at the level of admissible support sets, not at the level of the constructions carried by those sets.

5 Minimal Nontrivial Instance

The relevant notion of minimality is taken within the class of nonconstant finite typed Boolean functions with proper determinational support.

Let

$$f : \mathbf{2}^N \longrightarrow \mathbf{2}$$

be nonconstant and suppose that

$$R_f \subsetneq N.$$

Since f is nonconstant,

$$R_f \neq \varnothing.$$

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Since the determinational support is proper,

$$N \setminus R_f \neq \varnothing.$$

Thus the formal frame contains at least one real coordinate and at least one fictitious coordinate. Consequently,

$$\boxed{|N| \geq 2.}$$

If this ambient-frame lower bound is attained, so that

$$|N| = 2,$$

then necessarily

$$|R_f| = 1, \qquad |N \setminus R_f| = 1.$$

It follows that there is no intermediate support strictly between R_f and N, and hence

$$[R_f, N]_{\subseteq} = \{R_f, N\}.$$

Variant 11 attains the ambient-frame lower bound. Indeed,

$$N = \{p, q\}, \qquad R_{f_{11}} = \{q\},$$

so that

$$|N| = 2, \qquad |R_{f_{11}}| = 1, \qquad |N \setminus R_{f_{11}}| = 1.$$

Therefore,

$$\boxed{\text{Variant 11 is a minimal nontrivial instance of the common support interval}}$$

within the class of nonconstant finite typed Boolean functions with proper determinational support.

This minimality concerns the size of the ambient frame. It must be distinguished from the separate minimality of the interval itself. Whenever

$$|N \setminus R_f| = 1,$$

one has

$$[R_f, N]_{\subseteq} = \{R_f, N\},$$

so the common support interval contains exactly two elements. Such a two-element interval may, however, occur in a larger formal frame. Thus the cardinality of the interval alone does not establish minimality of the ambient frame.

No uniqueness claim is made.

6 Conclusion

Variant 11 is given by a finite typed Boolean function with

$$R_{f_{11}} = \{q\} \subsetneq N = \{p, q\}.$$

III-A subsequently establishes the complete family of exact syntactic supports:

$$\mathfrak{S}_N(f_{11}) = [\{q\}, \{p, q\}]_{\subseteq}.$$

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The Common Support Interval result then gives

$$\mathfrak{F}_N(f_{11}) = \mathfrak{S}_N(f_{11}).$$

Therefore the structural position of Variant 11 can now be stated precisely:

$$\boxed{\mathfrak{F}_N(f_{11}) = \mathfrak{S}_N(f_{11}) = \Big\{\{q\}, \{p, q\}\Big\}.}$$

Within the class of nonconstant finite typed Boolean functions with proper determinational support, Variant 11 attains the smallest possible ambient-frame size. It is therefore a minimal nontrivial instance of the common support interval.

The note ends at this structural identification.

References

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